
Global Representations Prevent Loss of Plasticity in Online Continual Learning

Anonymous Authors

Abstract

Continual learning in fully online settings is challenging due to non-stationary data streams and single-pass training without task boundaries or data reuse. In this regime, models often suffer from a *loss of plasticity*, i.e., a reduced ability to adapt to new data over time, leading to degradation in forward performance.

We propose Query-Only Attention (QOA), derived by simplifying attention-based models for the online continual learning setting, which learns a direct similarity function between queries and support examples without relying on keys or values. We show that this formulation enforces a consistent matching behavior, where inputs with similar outputs are mapped to similar representations. We provide a theoretical perspective supporting convergence toward a shared global solution across tasks, which stabilizes adaptation and mitigates loss of plasticity.

We evaluate QOA on long-horizon online versions of CIFAR-100, ImageNet, and additional benchmarks. Existing continual learning approaches, including replay, regularization, and re-initialization methods, often exhibit degraded forward performance under strict online, non-stationary settings due to slow adaptation. In contrast, QOA achieves higher sequential accuracy and near-zero cumulative regret, indicating faster and more reliable online adaptation.

Our focus is on forward performance and the preservation of plasticity in fully online settings, where adaptation must occur without task boundaries or data reuse.

1 Introduction

Continual learning remains a fundamental challenge in deep learning [37], where models must learn from non-stationary data streams without succumbing to catastrophic forgetting [19] or loss of plasticity [12]. In fully online settings, where data is observed only once and task boundaries are absent, models must continuously adapt to evolving distributions while maintaining strong forward performance.

While existing approaches mitigate forgetting using replay buffers [40] or regularization [19], preserving plasticity—the ability to acquire new knowledge over time—remains less understood. Moreover, methods designed to preserve plasticity often exhibit degraded performance in strict online, non-stationary settings.

Recent work suggests that attention mechanisms [34, 29, 17] can support stable adaptation across tasks [36]. Motivated by this, we ask: can the core mechanism of attention be simplified while retaining its ability to support continual adaptation? Further, what properties of attention enable it to mitigate loss of plasticity and achieve strong performance in online settings?

We propose Query-Only Attention (QOA), a simplified attention-based model that removes keys and values and instead learns a direct similarity function between queries and support examples. Alongside the model, we develop a simple theoretical perspective and empirical analysis that explain how such similarity-based interactions lead to stable adaptation over time. In particular, we show that this formulation enforces a consistent matching behavior, leading to convergence toward a shared

global solution across tasks. This behavior stabilizes learning and mitigates loss of plasticity over long horizons.

Our analysis shows that QOA exhibits stable task-adapted representations and reduced parameter drift across tasks. Additional comparisons with Model-Agnostic Meta-Learning [14], (Appendix) further highlight differences in adaptation behavior under online settings.

Key contributions:

- We derive *Query-Only Attention* (QOA) from the standard attention mechanism [34] for the online continual learning setting. By removing keys and learned value vectors and instead using direct query–support similarity with support-label aggregation, QOA preserves the support-based structure of attention while simplifying the prediction rule.
- We provide a theoretical analysis showing why both full attention and query-based continual learning can mitigate loss of plasticity: their objectives induce a consistent query–support matching behavior across tasks. This leads to convergence toward a shared *global* solution rather than repeatedly adapting to task-specific local solutions.
- We show that QOA is computationally better suited to online continual learning than full attention. While full attention requires pairwise buffer interactions, QOA only computes query–support interactions needed for the current prediction, reducing the dependence on buffer size from quadratic to linear.
- We conduct a comprehensive empirical comparison in difficult long-horizon, non-stationary online settings against replay-, regularization-, activation-, and re-initialization-based continual learning baselines. QOA achieves superior prequential accuracy, showing stronger online performance and sustained plasticity.
- We evaluate convergence speed using relative cumulative regret. QOA achieves near-zero regret on several classification benchmarks, indicating faster convergence and more reliable adaptation compared to existing continual learning methods.
- We introduce a task-wise augmentation protocol for large-scale benchmarks (CIFAR-100, Tiny ImageNet, and ImageNet-1000), where each task is generated via deterministic, per-task augmentations. This induces controlled distribution shifts across tasks, enabling realistic evaluation of online adaptation and plasticity without relying on repeated data or explicit task boundaries.
- As supporting analysis, we include Hessian rank evaluation [25] and comparisons with gradient-based meta-learning methods [14], which provide additional evidence for the role of global solution convergence in preserving plasticity (Appendix).

2 Related Work

Deep neural networks typically assume stationary data and multiple passes over the dataset, whereas online continual learning operates on non-stationary streams where each sample is observed once. This setting introduces two fundamental challenges: *catastrophic forgetting*, the degradation of previously learned tasks [28, 18, 30], and *loss of plasticity*, the reduced ability to learn new tasks over time [12, 2, 9].

Several recent methods explicitly target loss of plasticity. Continual Backpropagation [12] restores plasticity through selective re-initialization, Concat ReLU [2] improves representational capacity through activation design, and regenerative regularization [1] encourages parameters to remain adaptable. Regularization methods such as EWC [19], SI [39], and LwF [26], as well as replay methods such as ER [31] and DER [5], are also widely used to stabilize continual learning.

However, many of these methods are evaluated in relatively controlled settings, often with clear task boundaries or evaluation focused on end-of-task accuracy. End-of-task evaluation can overstate adaptability because it measures performance after the model has already trained on the current task. In strict online settings with distribution shifts across tasks, the more relevant question is whether the model adapts quickly as data arrives. We therefore emphasize prequential accuracy and cumulative regret, which directly measure online adaptability and convergence speed.

Recent work suggests that attention-based models can mitigate loss of plasticity in controlled or toy settings [36]. However, the mechanisms underlying this behavior are not well understood. In this

work, we provide a simple theoretical perspective, supported by empirical analysis, explaining why similarity-based interactions in attention help preserve plasticity.

Building on this insight, we derive Query-Only Attention (QOA), a simplified formulation tailored for online continual learning. QOA retains the query–support matching mechanism of attention while removing keys and learned value vectors, reducing computational cost from $\mathcal{O}(n^2d)$ to $\mathcal{O}(nd)$. We show that both full attention and query-based models can converge toward a shared global solution across tasks, which stabilizes learning and mitigates loss of plasticity. Empirically, QOA achieves improved cumulative regret and prequential accuracy in more realistic, non-stationary settings, demonstrating stronger online performance under distribution shift.

More broadly, our approach connects attention mechanisms, in-context learning [13, 38, 3], and meta-learning frameworks such as MAML [14] and metric-based methods [35, 20, 33]. Finally, to study plasticity dynamics, we adopt curvature-based analyses such as effective Hessian rank [25, 32], where stable rank indicates sustained adaptability.

3 Problem Statement

We consider an *online continual learning* setting where a model observes a stream of data sequentially, each sample seen once (no epochs or replay) [6, 4]. The stream is generated from tasks $\{T_1, T_2, \dots\}$, where each task T_i follows a (potentially non-stationary) distribution $\mathcal{D}_i(x, y)$.

The model is trained online without access to task identities or boundaries, updating parameters using only current observations (single samples or small batches) [21, 8, 4].

Evaluation is performed in a forward (prequential) manner along the stream, where each sample is first used for testing before training, reflecting real-world deployment scenarios [10, 15, 6, 27].

We measure **forward performance (plasticity)** via predictive accuracy on incoming samples prior to training, where degradation over time indicates loss of plasticity.

4 Method

4.1 From Attention to Query-Only Attention

We derive Query-Only Attention (QOA) from standard attention in the online continual learning setting. Let the current input at time t be

$$x_t \in \mathbb{R}^{1 \times d},$$

and let the replay buffer contain m support examples

$$B_t = \{(x_j, y_j)\}_{j=1}^m, \quad x_j \in \mathbb{R}^{1 \times d}.$$

For classification, y_j denotes the label of support example j ; for regression, y_j denotes the target value.

At time t , full attention would stack the current query and the buffer elements into

$$X_t = \begin{bmatrix} x_t \\ x_1 \\ \vdots \\ x_m \end{bmatrix} \in \mathbb{R}^{(m+1) \times d}.$$

It then computes

$$Q = X_t W_Q, \quad K = X_t W_K, \quad V = X_t W_V,$$

where

$$W_Q, W_K, W_V \in \mathbb{R}^{d \times d}, \quad Q, K, V \in \mathbb{R}^{(m+1) \times d}.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) \in \mathbb{R}^{(m+1) \times (m+1)},$$

and the output is

$$O = AV \in \mathbb{R}^{(m+1) \times d}.$$

This computes an output representation for the current query and for every support element in the buffer. However, in online continual learning, we only need the prediction for the current query x_t . Therefore, computing all pairwise interactions among the $m + 1$ elements is unnecessary.

The full attention score computation has cost

$$QK^\top : \mathbb{R}^{(m+1) \times d} \mathbb{R}^{d \times (m+1)} \rightarrow \mathbb{R}^{(m+1) \times (m+1)},$$

which scales as

$$\mathcal{O}(m^2 d).$$

The value aggregation step

$$AV : \mathbb{R}^{(m+1) \times (m+1)} \mathbb{R}^{(m+1) \times d} \rightarrow \mathbb{R}^{(m+1) \times d}$$

also scales as $\mathcal{O}(m^2 d)$. Thus, full attention has quadratic dependence on the support size.

For predicting the current query, the only required interactions are between x_t and each support example x_j . In standard attention, this interaction is given by

$$q_t k_j^\top = (x_t W_Q)(x_j W_K)^\top,$$

where

$$q_t = x_t W_Q \in \mathbb{R}^{1 \times d}, \quad k_j = x_j W_K \in \mathbb{R}^{1 \times d}.$$

This term is a learned similarity between the current query and support example j .

QOA replaces this projected dot-product similarity with a direct learned similarity function:

$$s_\theta(x_t, x_j) = g_\theta([x_t, x_j]),$$

where

$$[x_t, x_j] \in \mathbb{R}^{1 \times 2d}$$

is the concatenation of the query and support input, and

$$g_\theta : \mathbb{R}^{2d} \rightarrow \mathbb{R}$$

is a learned scoring function. Computing this score for all support examples gives

$$s_t = (s_\theta(x_t, x_1), \dots, s_\theta(x_t, x_m)) \in \mathbb{R}^m.$$

We normalize the scores across the support set:

$$\alpha_{t,j} = \frac{\exp(s_\theta(x_t, x_j))}{\sum_{k=1}^m \exp(s_\theta(x_t, x_k))}, \quad \alpha_t \in \mathbb{R}^m.$$

In standard attention, the output is a weighted sum of learned value vectors. In online continual learning, each support example already has an associated label or target. Therefore, QOA replaces learned value vectors with support labels. For classification, the prediction is

$$\hat{y}_t = \sum_{j=1}^m \alpha_{t,j} e(y_j),$$

where $e(y_j) \in \mathbb{R}^C$ is the one-hot encoding of the support label and C is the number of classes. Hence,

$$\hat{y}_t \in \mathbb{R}^C.$$

For regression, the one-hot label vector is replaced by the support target:

$$\hat{y}_t = \sum_{j=1}^m \alpha_{t,j} y_j.$$

Thus, QOA keeps only the query–support interactions required for the current prediction. It removes full pairwise buffer attention, explicit key projections, and learned value vectors. If g_θ is an MLP with hidden size h , computing all m query–support scores costs

$$\mathcal{O}(mdh).$$

When h is fixed, this becomes

$$\mathcal{O}(md).$$

In contrast, full attention scales as

$$\mathcal{O}(m^2 d).$$

Therefore, QOA reduces the dependence on support size from quadratic to linear.

Training Procedure. The model is trained in a fully online manner. At each step, the current sample is added to the buffer using a FIFO policy, a support set is sampled, and the model is updated via gradient descent on the prediction loss.

Algorithm 1 Query-Only Attention (QOA) with Replay Buffer

Input: Data stream (x_t, y_t) ; buffer $\mathcal{B}$ of size n ; support size m ; scoring network g_θ

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1: Initialize parameters  $\theta$ ; buffer  $\mathcal{B} \leftarrow \emptyset$ 
2: for each timestep  $t$  with sample  $(x_t, y_t)$  do
3:    $\mathcal{S} = \{(x_j, y_j)\}_{j=1}^m \subset \mathcal{B}$ 
4:   Vectorized query–support scoring:
5:   Stack supports:  $X_S \in \mathbb{R}^{m \times d}$ 
6:   Repeat query:  $X_t^{rep} \in \mathbb{R}^{m \times d}$ 
7:   Concatenate:  $Z = [X_t^{rep}, X_S] \in \mathbb{R}^{m \times 2d}$ 
8:   Scores:  $s_t = g_\theta(Z) \in \mathbb{R}^m$ 
9:   Aggregate prediction:
10:  if classification then
11:     $\hat{y}_t = s_t^\top Y_S$ , where  $Y_S \in \mathbb{R}^{m \times C}$ 
12:  else
13:     $\hat{y}_t = s_t^\top y_S$ 
14:  end if
15:  Loss and update:
16:   $\mathcal{L}_t \leftarrow \ell(\hat{y}_t, y_t)$ 
17:   $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_t$ 
18:  Update buffer (after learning):
19:  Insert  $(x_t, y_t)$  into  $\mathcal{B}$ ; evict oldest if  $|\mathcal{B}| > n$ 
20: end for
```

Note: The current sample (x_t, y_t) is not included in the support set. The buffer is updated only after the parameter update to avoid trivial query–support matching.

5 Experiments

5.1 Experimental Setup

We evaluate our method on five benchmarks: Slowly Changing Regression (SCR), Permuted MNIST, CIFAR-100, Tiny ImageNet, and ImageNet, under a strict online continual learning setting.

General protocol. Data is presented as a continuous stream without task identifiers, and each sample is observed only once. Tasks are constructed to exhibit both *distributional shift* and *recurring classes*, creating a challenging non-stationary environment. Unlike standard continual learning benchmarks that rely on task boundaries or multiple epochs over data, our setting enforces single-pass learning with evolving input distributions.

CIFAR-100, Tiny ImageNet, and ImageNet. We use the CIFAR-100 dataset [22], the Tiny ImageNet dataset [23], and ImageNet [11]. For CIFAR-100 and Tiny ImageNet, each task consists of 5 classes sampled uniformly at random (with replacement) from the global label set. To introduce distributional shift, we apply a fixed augmentation per task (e.g., cropping, rotation, or color transformations) to all images in that task. This design induces two key challenges: (i) *recurring classes*, where the same class reappears across tasks, and (ii) *distributional variation*, where the input distribution for the same class changes across tasks due to augmentations. As a result, the model must continuously adapt to new variations of previously seen classes without explicit task information.

For ImageNet, we adopt a similar setup but construct binary classification tasks. Each task consists of 2 classes sampled from the 1000-class dataset, with task-specific augmentations applied as above. This setting scales the problem to a large and diverse dataset while retaining the challenges of recurring classes and distribution shift.

Permuted MNIST. Each task consists of all 10 classes, with a task-specific random permutation applied to the input pixels. This results in tasks that share the same label space but differ in input

distribution, requiring the model to adapt to changing representations while maintaining classification performance [16].

Slowly Changing Regression (SCR). We use the SCR benchmark from [12], where the data distribution evolves gradually over time. This setting isolates the effect of non-stationary and is specifically designed to study loss of plasticity.

Evaluation Protocol and Metrics. Data is presented as a continuous stream without task identifiers, and each sample is observed only once. Tasks are constructed to exhibit both *distributional shift* and *recurring classes*, creating a challenging non-stationary environment. Unlike standard continual learning benchmarks that rely on task boundaries or multiple epochs over data, our setting enforces single-pass learning with evolving input distributions.

We evaluate models using *prequential accuracy*, where each sample is first used for testing before the model is updated [10]:

$$\text{Acc}_T^{(m)} = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(\hat{y}_t^{(m)} = y_t),$$

where $\hat{y}_t^{(m)}$ is the prediction of method m at time t . Under this protocol, performance on a new task is initially low and improves as more samples are observed. As a result, prequential accuracy is typically lower than evaluation based on held-out test sets measured after training on a task. This evaluation more closely reflects real-world deployment, where predictions must be made before adaptation, and explicitly penalizes slow adaptation since models incur errors over the early portion of each task.

5.2 Experiment Results

Method	PMNIST		CIFAR-100		TinyIN		ImageNet		SCR	
	Start $\uparrow$	End $\uparrow$	Start $\uparrow$	End $\uparrow$	Start $\uparrow$	End $\uparrow$	Start $\uparrow$	End $\uparrow$	Start $\downarrow$	End $\downarrow$
Vanilla BP	65.42	53.44	27.69	21.09	22.17	20.52	52.73	52.41	0.970	0.641
CBP	62.58	67.09	38.84	51.57	29.87	35.10	61.68	55.74	1.094	0.693
Concat ReLU	74.72	78.98	28.54	44.15	23.53	25.21	54.09	55.10	0.814	0.647
Dark Exp.	73.88	64.15	28.28	33.59	23.27	25.66	53.64	59.25	0.961	0.447
ER	71.38	64.44	30.49	24.00	23.71	21.37	52.78	59.19	0.995	0.512
EWC	49.84	39.09	27.86	65.17	22.45	45.47	53.07	59.44	2.761	2.897
Full Attention	29.70	88.66	37.51	84.35	29.22	62.25	59.57	76.16	1.118	0.794
QOA (Ours)	80.86	87.81	47.55	88.88	36.59	67.78	65.98	85.97	7.542	0.510
Reg. Reg.	66.80	68.13	27.05	21.93	22.38	19.43	52.70	52.44	1.006	0.701

Table 1: Prequential performance across datasets. Start and End denote early-stage and long-horizon performance, respectively. Accuracy metrics ($\uparrow$) are higher-is-better, while SCR loss ($\downarrow$) is lower-is-better.

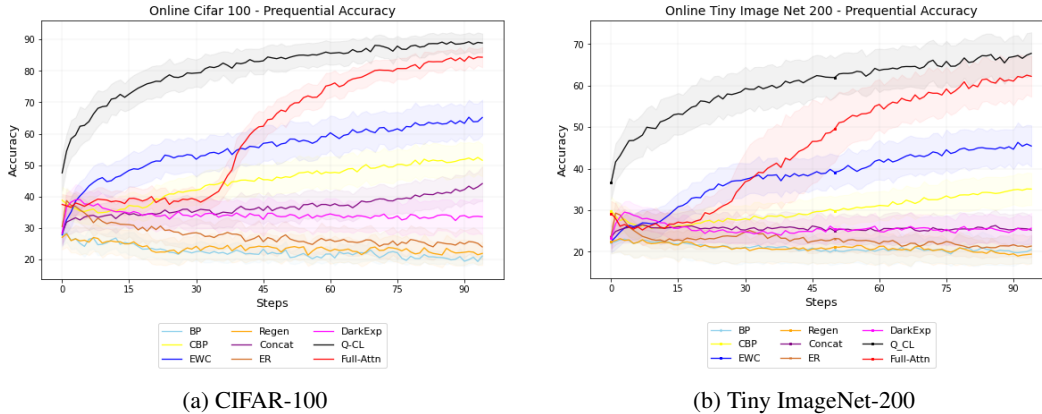


Figure 1: Prequential accuracy over 10⁴ tasks. Each point is averaged over 100 tasks. QOA achieves consistently higher accuracy and faster adaptation compared to baselines.

We present prequential accuracy plots for CIFAR-100 and Tiny ImageNet in the main text. Corresponding results for all other datasets, along with forward performance plots, are deferred to the Appendix due to space limitations.

Observations on the performance Across all benchmarks, **query-only attention** consistently outperforms all baselines, with full-attention models as the closest competitor. However, query-only attention converges faster and scales more effectively on harder tasks. It mitigates both loss of plasticity and forgetting, highlighting its practicality under limited compute and memory constraints.

These results suggest that preserving plasticity directly improves the reliability and interpretability of model updates, enabling safer adaptation under non-stationary data distributions. Importantly, under the prequential evaluation protocol, several methods designed to address loss of plasticity still achieve poor performance due to slow adaptation, which is explicitly penalized. In contrast, query-only attention maintains both rapid adaptability and strong long-term performance.

5.2.1 Regret

To quantify convergence speed, we use *relative cumulative regret* following standard online learning formulations [7]. Let $\ell_t^{(m)}$ denote the loss of method m at time t , and define the instantaneous best loss as $\ell_t^* = \min_{m'} \ell_t^{(m')}$. The regret of method m over T steps is then

$$R_T^{(m)} = \sum_{t=1}^T (\ell_t^{(m)} - \ell_t^*).$$

Lower regret indicates faster adaptation and improved online performance. For classification, $\ell_t^{(m)}$ corresponds to cross-entropy loss, while for regression we use mean squared error. We additionally report the *adaptation gap*

$$\Delta_t = A_t^{\text{forward}} - A_t^{\text{preq}},$$

which measures within-task improvement and serves as a diagnostic of adaptation speed. (*Regret is defined relative to the set of evaluated methods.*)

Method	PMNIST ↓	CIFAR-100 ↓	TinyIN ↓	ImageNet ↓	SCR ↓
Vanilla BP	2859.33	5547.38	3703.83	1445.60	10.08
CBP	2027.32	3427.54	2841.49	1247.24	12.10
Concat ReLU	839.95	4177.70	3265.27	1348.52	8.97
Dark Exp.	1905.97	4422.49	3268.27	1171.60	1.75
ER	1710.34	5042.10	3521.32	1065.16	4.70
EWC	4393.06	2430.45	2070.90	1250.68	118.07
Full Attention	435.69	1933.06	1409.17	1072.36	22.09
QOA (Ours)	22.34	0.00	0.00	0.00	10.89
Reg. Reg.	1877.02	5432.98	3705.72	1443.58	13.50

5.2.2 Regret analysis.

Relative cumulative regret across datasets (lower is better). QOA consistently achieves the lowest regret on classification benchmarks, often approaching zero, indicating rapid adaptation to the data stream. In contrast, replay- and regularization-based methods exhibit substantially higher regret, reflecting slower convergence under non-stationary conditions. On SCR, the best performance is achieved by Dark Experience, suggesting distinct optimization dynamics in regression settings.

Global solution intuition. In QOA, predictions are formed via similarity-weighted aggregation over a support set. Under cross-entropy training, the model is encouraged to assign higher similarity to supports with matching labels than to non-matching ones.

This induces a task-invariant structural constraint on the similarity function $s_\theta(\cdot, \cdot)$ that holds across all tasks. As a result, the task-level representations

$$\theta'_t = [s_\theta(x_t, x_1), \dots, s_\theta(x_t, x_m)]$$

tend to align across tasks:

$$\theta'_t \approx \theta'_s \quad \forall t, s.$$

We refer to this regime as a global solution, where a shared relational structure explains all tasks, leading to stable learning dynamics and reduced loss of plasticity. A detailed derivation of this global solution is provided in Appendix B.

Interpretation. Under this view:

- datapoints $\rightarrow$ share structure $\rightarrow$ learn θ
- tasks $\rightarrow$ share structure $\rightarrow$ learn θ'_t

Thus, learning across tasks reduces to discovering a shared task-level representation. Empirically, convergence of θ'_t is associated with sustained performance and mitigation of loss of plasticity.

Comparison to gradient-based adaptation. Gradient-based meta-learning methods such as MAML compute task-specific parameters

$$\theta'_t = \theta - \alpha \nabla_{\theta} \mathcal{L}_t(\theta), \quad (1)$$

where $\mathcal{L}_t$ depends on the current task.

Unlike query-based or attention models, MAML does not induce a task-invariant structural constraint on the relationship between inputs. The adaptation step is driven entirely by task-specific gradients, which vary across tasks. As a result, the resulting representations θ'_t are not required to align or converge across tasks.

In contrast, our formulation induces a consistent matching structure across tasks through the learning objective, enabling convergence to a shared task-level representation.

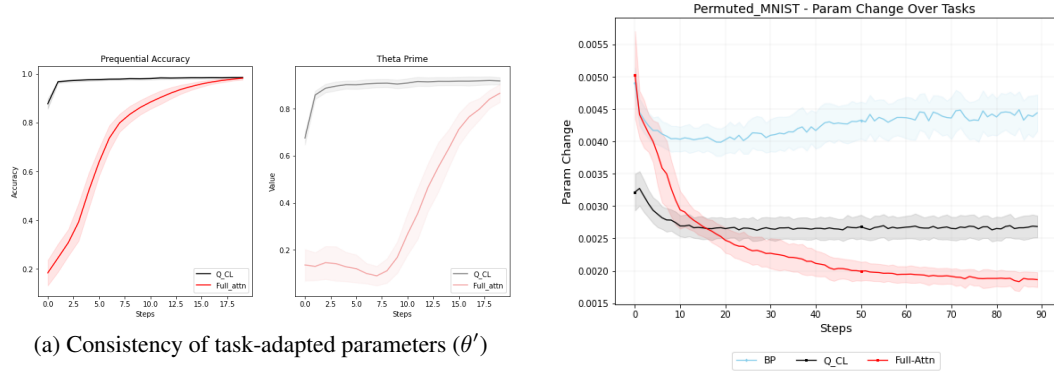


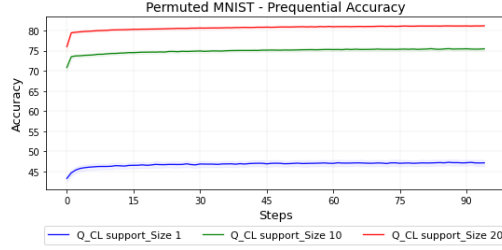
Figure 2: Model dynamics across tasks. QOA exhibits stable task-adapted representations and reduced parameter drift.

5.3 Ablation Study: Effect of Buffer Size

We study the impact of buffer size on performance by varying the number of support samples n while keeping all other factors fixed.

As shown in Figure 3, performance improves consistently as the buffer size increases, indicating that the model effectively leverages additional past information. The gains saturate beyond a certain point, suggesting diminishing returns for very large buffers.

This behavior is consistent with the linear scaling of our method with respect to buffer size, enabling efficient utilization of larger buffers.



(a) Ablation study on Permuted MNIST

Figure 3: Prequential accuracy over 10^4 tasks. Each point is averaged over 100 tasks. The accuracy improves as the support size increases.

6 Conclusion

We introduced a Query-Only Attention method for continual learning. Our results show that it reduces loss of plasticity even when task boundaries are unknown, while outperforming existing methods without requiring repeated training on past data. When task boundaries are available, it can also reduce catastrophic forgetting. In addition, it is more computationally efficient than full attention.

We analyzed the method by connecting it to MAML and full attention, and by interpreting it through the idea of global versus local solutions. We also showed its relationship to k -nearest neighbor methods. These connections were supported by experiments on multiple benchmarks. Our Hessian-based analysis further suggests that maintaining curvature helps preserve plasticity during training.

A limitation of our approach is the reliance on a support set, which makes handling catastrophic forgetting more difficult in some settings. In future work, we plan to evaluate larger and more diverse datasets, develop stronger theoretical understanding, and reduce dependence on explicit support sets.

Code Availability

The implementation of Query-Only Attention and all experiments are available at: <https://github.com/gauthambekal93/query-only-attention-for-continual-learning>.

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A Technical Appendices and Supplementary Material

A.1 Broader impacts

This work is primarily foundational research in continual learning. The proposed query-only attention mechanism and accompanying analysis aim to improve the understanding of plasticity and forgetting in neural networks. Potential positive impacts include enabling more efficient and adaptive AI systems, which could reduce retraining costs, improve energy efficiency, and support applications such as robotics, healthcare monitoring, and lifelong personal assistants.

At the same time, continual learning technologies can be misused in domains such as surveillance or profiling, where adaptive models might amplify privacy concerns or biases. While the present work is not directly deployable, these risks highlight the need for responsible use and safeguards in future applications.

Overall, this research contributes theoretical and empirical insights into the foundations of continual learning, with the aim of advancing the field in a transparent and beneficial direction.

A.2 Limitations

Our work has a few important limitations. First, the query-only attention model relies on a support set, which can be restrictive for mitigating catastrophic forgetting in practical continual learning scenarios. Second, our evaluation is limited to two benchmarks (Permuted MNIST and Slowly Changing Regression); broader validation on more complex datasets is needed to confirm generality. Third, while we provide theoretical analysis and intuition linking query-only attention to MAML and k NN, we do not include formal proofs. We acknowledge these as areas for future work, particularly in extending the experimental scope and strengthening the theoretical foundation.

B Task-Level Representations and Derivation of Global Solutions

From datapoints to tasks. In standard supervised learning, the objective is:

$$\min_{\theta} \sum_{t=1}^T \sum_{i=1}^{n_t} \mathcal{L}(f_{\theta}(x_{t,i}), y_{t,i}), \quad (2)$$

where $x_{t,i}$ and $y_{t,i}$ denote the i -th datapoint and label in task t , and θ are model parameters. Learning succeeds when datapoints share a common structure that can be captured by a single parameter vector θ .

Support-based prediction. In query-based and attention models, predictions are computed using a buffer

$$B_t = \{(x_j, y_j)\}_{j=1}^{|B_t|}.$$

For a query $x_{t,i}$, the prediction takes the form:

$$f_{\theta}(x_{t,i}, B_t) = \sum_{j=1}^{|B_t|} \alpha_{t,i,j} V(y_j), \quad (3)$$

where:

- $\alpha_{t,i,j}$ are normalized similarity weights,
- $\alpha_{t,i,j} \propto \exp(s_{\theta}(x_{t,i}, x_j))$,
- $s_{\theta}(x_{t,i}, x_j)$ is the learned similarity function,
- $V(y_j)$ denotes the value associated with support j (e.g., $e(y_j)$ for classification or y_j for regression).

The buffer is constructed online (FIFO) and is not conditioned on task identity.

Task-level representation. We define the task-level representation induced by the model as:

$$\theta'_t := [s_\theta(x_t, x_1), \dots, s_\theta(x_t, x_{|B_t|})] \in \mathbb{R}^{|B_t|}, \quad (4)$$

which captures how the model relates a query from task t to its support set.

Shared structure induced by the objective. Consider a query (x_q, y_q) and a support set

$$B_t = \{(x_j, y_j)\}_{j=1}^{|B_t|}.$$

In our model, predictions are formed as a similarity-weighted aggregation of support labels. When trained with a cross-entropy loss, the objective encourages the model to assign high probability to the correct label y_q .

This implicitly induces the following behavior: supports with matching labels ($y_j = y_q$) must receive higher similarity scores than supports with non-matching labels ($y_j \neq y_q$).

Formally, defining

$$\mathcal{S}^+(q) = \{j \mid y_j = y_q\}, \quad \mathcal{S}^-(q) = \{j \mid y_j \neq y_q\},$$

the training objective encourages:

$$\mathbb{E}_{j \in \mathcal{S}^+(q)} s_\theta(x_q, x_j) > \mathbb{E}_{j \in \mathcal{S}^-(q)} s_\theta(x_q, x_j), \quad \forall t. \quad (5)$$

Thus, this is not an explicit constraint imposed on the model, but rather a structural property induced by the learning objective, and remains consistent across tasks.

Global solution (key idea). In standard learning, a single parameter θ is sufficient when all datapoints follow a shared pattern. Here, we extend this idea by treating each *task* as a higher-level datapoint.

The representation θ'_t summarizes the structure of task t . If tasks share a common pattern (as defined by the constraint above), then the induced representations θ'_t should converge to a common value:

$$\theta'_t \approx \theta'_s \quad \forall t, s. \quad (6)$$

We refer to this regime as a *global solution*, where a single task-level representation explains all tasks. In this case, the model does not need to continually shift its representation, leading to stable learning dynamics.

Interpretation. Under this view:

- datapoints $\rightarrow$ share structure $\rightarrow$ learn θ
- tasks $\rightarrow$ share structure $\rightarrow$ learn θ'_t

Thus, learning across tasks reduces to discovering a shared task-level representation. Empirically, convergence of θ'_t is associated with sustained performance and mitigation of loss of plasticity.

Comparison to gradient-based adaptation. Gradient-based meta-learning methods such as MAML compute task-specific parameters

$$\theta'_t = \theta - \alpha \nabla_\theta \mathcal{L}_t(\theta), \quad (7)$$

where $\mathcal{L}_t$ depends on the current task.

Unlike query-based or attention models, MAML does not induce a task-invariant structural constraint on the relationship between inputs. The adaptation step is driven entirely by task-specific gradients, which vary across tasks. As a result, the resulting representations θ'_t are not required to align or converge across tasks.

In contrast, our formulation induces a consistent matching structure across tasks through the learning objective, enabling convergence to a shared task-level representation.

C Additional Results

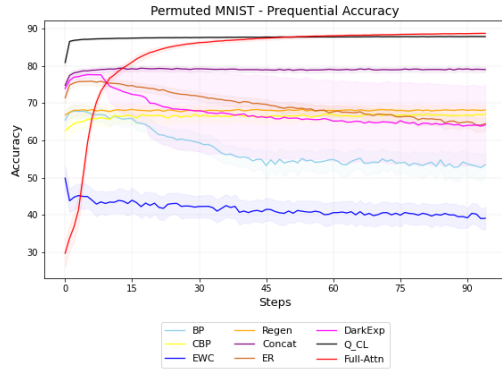


Figure 4: PMNIST (Preq.)

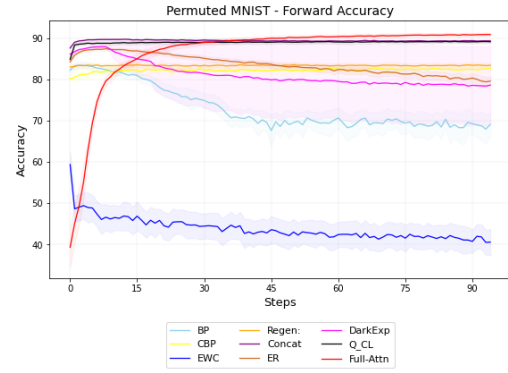


Figure 5: PMNIST (Fwd.)

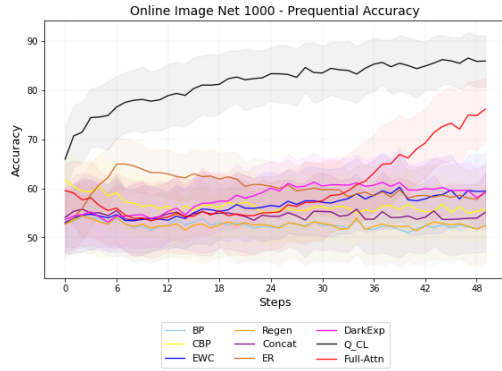


Figure 6: ImageNet (Preq.)

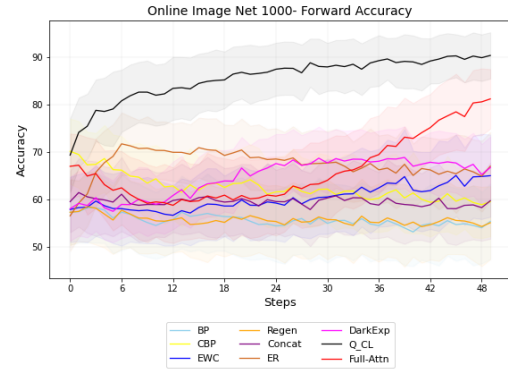


Figure 7: ImageNet (Fwd.)

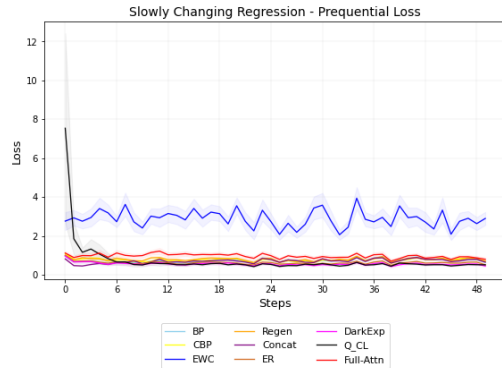


Figure 8: SCR (Preq.)

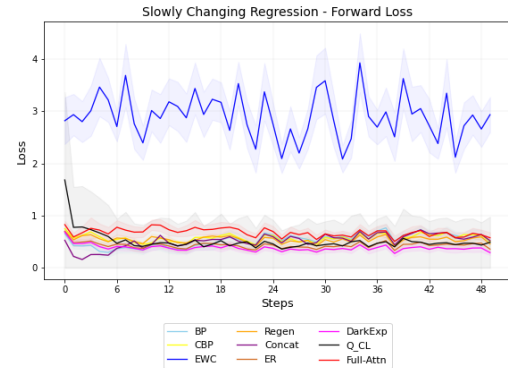


Figure 9: SCR (Fwd.)

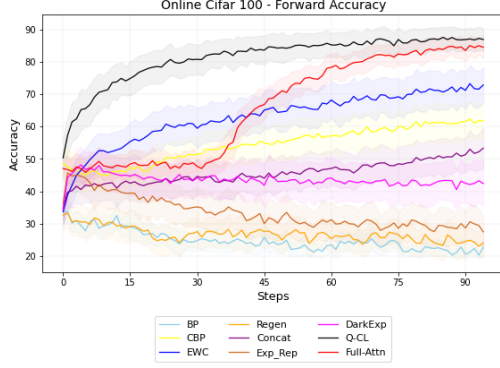


Figure 10: CIFAR-100 (Fwd.)

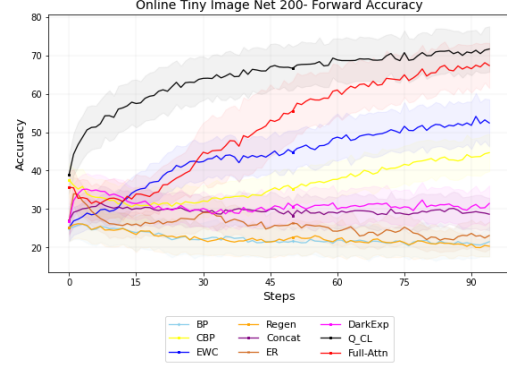


Figure 11: TinyIN (Fwd.)

Figure 12: Prequential (Preq.) and forward (Fwd.) performance across datasets. Prequential curves evaluate performance on each sample before model update, capturing online adaptation dynamics, while forward curves measure performance after adaptation within each task. Across classification benchmarks, QOA exhibits consistently higher prequential accuracy and faster convergence, indicating strong plasticity and rapid adaptation. On SCR, performance differences highlight distinct optimization behavior in regression settings.

Discussion. Prequential performance reflects the model’s ability to adapt to the data stream, while forward performance captures its final accuracy after within-task updates. Across all classification datasets, QOA achieves higher prequential accuracy, indicating faster adaptation and reduced loss of plasticity. This is further supported by near-zero regret values in Table 5.2.1. In contrast, replay and regularization-based methods exhibit slower adaptation, as reflected by lower prequential performance despite competitive forward accuracy. On SCR, the behavior differs, suggesting that regression settings induce distinct optimization dynamics compared to classification tasks.

C.1 MAML with replay buffer

Note. The adaptation of MAML with replay buffer is still work in progress. We include it here to illustrate a potential direction for combining meta-learning with large buffers in continual learning, but do not claim it as a finalized or fully validated algorithm. That said, our preliminary results are promising for one of the experiments, and suggest this variant may provide a complementary approach to attention-based or query-only models.

We adapt MAML to the continual learning setting by introducing a large replay buffer. To minimize interference across tasks, the buffer is evenly partitioned into t sub-buffers, one per sampled task. Without such partitioning, the sampled support and query examples from different tasks would overlap excessively, leading to degraded task separation and unstable meta-updates. Each task provides a small support set (s) and query set (q), which is sufficient for the MAML inner/outer loops. An important property of this setup is that the support/query sizes remain much smaller than those required by attention-based or query-only models. This makes MAML particularly well-suited for backward evaluation (catastrophic forgetting), since only a small set of examples per task is needed to adapt. Standard MAML inner-loop adaptation and outer-loop meta-updates are then applied on these partitioned tasks. The details of the algorithm are presented in the appendix section.

Algorithm 2: MAML with Replay Buffer (work in progress)

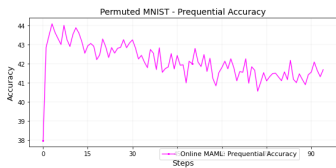


Figure 13: PMNIST (Preq.)

Input: Stream (x_t, y_t) ; replay buffer $\mathcal{B}$ of size N ; tasks t ; support s ; query q ; learning rates α, β

- 1: Initialize $\theta, \mathcal{B} \leftarrow \emptyset$
- 2: **for** each incoming (x_t, y_t) **do**
- 3: Insert (x_t, y_t) into $\mathcal{B}$; evict oldest if $|\mathcal{B}| > N$
- 4: Partition $\mathcal{B}$ into t equal sub-buffers
- 5: **for** each task $i = 1 \dots t$ **do**
- 6: Sample s support and q queries from sub-buffer i
- 7: Inner update on support with step size α
- 8: Outer update on queries with step size β
- 9: **end for**
- 10: **end for**

C.2 Hessian rank analysis

To study plasticity, we analyze the *effective rank* of the Hessian [25],[32], defined as

$$\text{erank}(H) = \exp\left(-\sum_{i=1}^n p_i \log p_i\right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}, \quad (8)$$

where $\{\lambda_i\}$ are the eigenvalues of the Hessian. A stable non-decreasing effective rank indicates models that preserve plasticity across tasks.

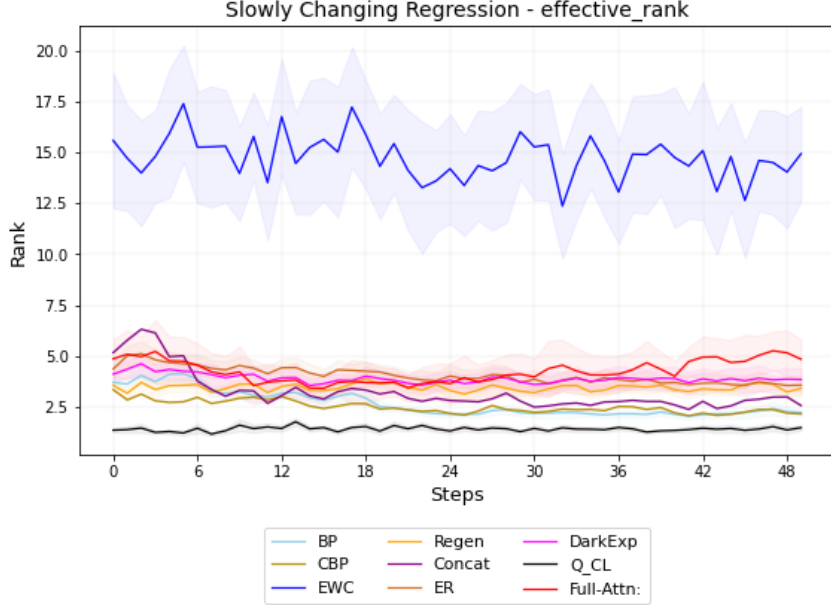


Figure 14: Effective rank dynamics across tasks. QOA maintains stable rank, indicating preserved plasticity. The effective rank provides a measure of model plasticity. We measure it at the start of each task using the Hessian of the final layer only, since full-Hessian computation is infeasible. We measure it only on Permuted MNIST and SCR and not on ImageNet due to computational constraints. Also, the effective rank has been normalized since it depends on network size. In both Permuted MNIST and SCR, vanilla backpropagation shows a steady drop in effective rank, aligning with loss of plasticity. All other models show minimal drop, indicating preserved plasticity.

D Implementation Details

Task-wise augmentations. To induce controlled non-stationarity in the data stream, we apply a fixed augmentation configuration per task for CIFAR-100, Tiny ImageNet, and ImageNet. For each task t , augmentation parameters are deterministically generated from the task index using a hash function, ensuring reproducibility across runs. These parameters define the strength of each transformation and remain fixed for all samples within the task.

The augmentation pipeline includes horizontal flipping, random cropping with reflection padding, additive Gaussian noise, spatial blurring via average pooling, brightness and contrast adjustments, channel-wise color shifts, desaturation toward grayscale, and cutout masking. While the transformation magnitudes are fixed per task, stochasticity is introduced at the sample level (e.g., random flip decisions, crop offsets, and noise realizations), resulting in diverse but task-consistent variations.

Formally, each task corresponds to a distribution $\mathcal{D}_t(x, y)$ obtained by applying a transformation $\mathcal{T}_t$ to the underlying dataset, where $\mathcal{T}_t$ is parameterized by task-specific values for crop size, noise level, blur kernel size, brightness, contrast, color shift, desaturation, and cutout region size. This setup ensures that the same classes reappear across tasks under different input distributions, creating a challenging scenario with recurring classes and evolving visual statistics.

The same task-specific transformation is applied consistently to both training and test samples within each task, ensuring that evaluation reflects adaptation to the current distribution rather than domain mismatch.

Augmentation parameters are generated as deterministic functions of the task index, avoiding additional sources of randomness across runs.

Hardware. All experiments were conducted on a single NVIDIA Ada-series GPU.

D.1 Common Implementation Details

Weights are initialized using Kaiming (He) uniform initialization, with appropriate gain for ReLU hidden layers and linear output layers, and biases are initialized to zero. All models are trained using the Adam optimizer with ReLU activations. Seeds 20, 30, 40 were used.

D.2 Data and Task Specification

Parameter	PMNIST	SCR	CIFAR-100	TinyIN	ImageNet
num_tasks	10000	–	10000	10000	5000
total_classes	10	–	100	200	1000
classes_per_task	10	–	5	5	2
num_old_task_window	5	5	5	5	2
num_datapoints_per_timestep	400	1	10	10	10
num_data_points	–	10M	–	–	–
train_size	–	9000	–	–	–
flip_after	–	10000	–	–	–
change_after	60000	–	–	–	–
image_size	7 x 7	–	3 x 32 x 32	3 x 32 x 32	3 x 128 x 128

Table 2: Implementation details for all benchmarks. PMNIST = Permuted MNIST, SCR = Slowly Changing Regression, CIFAR-100, TinyIN = Tiny ImageNet, ImageNet-1000.

D.2.1 Permuted MNIST: Model Specifications

Parameter	BP	CBP	EWC	Regen	Concat	ER	DarkExp	Full-Attn	QOA
input_size	49	49	49	49	49	49	49	49	49
num_features	200	200	200	200	200	200	200	100	40
hidden_layers	4	4	4	4	4	4	4	–	4
attention_layers	–	–	–	–	–	–	–	3	–
step_size	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4
weight_decay	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
maturity	–	100	–	–	–	–	–	–	–
decay_rate	–	0.99	–	–	–	–	–	–	–
utility_type	–	adaptable	–	–	–	–	–	–	–
replacement_rate	–	1e-4	–	–	–	–	–	–	–
ewc_lambda	–	–	100000	–	–	–	–	–	–
regenerative_lambda	–	–	–	0.01	–	–	–	–	–
buffer_size	–	–	–	–	–	2000	2000	1600	1600
buffer_type	–	–	–	–	–	reservoir	reservoir	–	–
samples_per_label	–	–	–	–	–	–	–	10	100
alpha	–	–	–	–	–	–	0.5	–	–
beta	–	–	–	–	–	–	0.5	–	–

Table 3: Model hyperparameters for Permuted MNIST. BP: Backpropagation, CBP: Continual Backprop, EWC: Elastic Weight Consolidation, Regen: Regenerative Regularization, Concat: Concat ReLU, ER: Experience Replay, DarkExp: Dark Experience Replay, Full-Attn: Full Attention, QOA: Query-Only Attention.

D.3 CIFAR-100 Model Specifications

Parameter	BP	CBP	EWC	Regen	Concat	ER	DarkExp	Full-Attn	QOA
step_size	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4
momentum	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
weight_decay	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
hidden_layer_size	100	100	–	–	100	100	100	100	50
optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
reset_head	–	false	–	–	–	–	–	–	–
early_stopping	–	true	–	–	–	–	–	–	–
replacement_rate	–	1e-5	–	–	–	–	–	–	–
utility_function	–	contribution	–	–	–	–	–	–	–
maturity_threshold	–	1000	–	–	–	–	–	–	–
noise_std	–	0	–	–	–	–	–	–	–
perturb_weights	–	false	–	–	–	–	–	–	–
ewc_lambda	–	–	1000	–	–	–	–	–	–
regenerative_lambda	–	–	–	1e-5	–	–	–	–	–
buffer_size	–	–	–	–	–	2000	2000	100	100
samples_from_buffer	–	–	–	–	–	10	10	–	–
fifo_samples_per_label	–	–	–	–	–	–	–	10	10
alpha	–	–	–	–	–	–	0.5	–	–
beta	–	–	–	–	–	–	0.5	–	–

Figure 15: CIFAR-100 model-specific hyperparameters for all methods.

D.4 Tiny ImageNet Model Specifications

Parameter	BP	CBP	EWC	Regen	Concat	ER	DarkExp	Full-Attn	QOA
step_size	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4
momentum	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
weight_decay	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
hidden_layer_size	100	100	–	–	100	100	100	100	50
optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
reset_head	–	false	–	–	–	–	–	–	–
early_stopping	–	true	–	–	–	–	–	–	–
replacement_rate	–	1e-5	–	–	–	–	–	–	–
utility_function	–	contribution	–	–	–	–	–	–	–
maturity_threshold	–	1000	–	–	–	–	–	–	–
noise_std	–	0	–	–	–	–	–	–	–
perturb_weights	–	false	–	–	–	–	–	–	–
ewc_lambda	–	–	1000	–	–	–	–	–	–
regenerative_lambda	–	–	–	1e-5	–	–	–	–	–
buffer_size	–	–	–	–	–	2000	2000	100	100
samples_from_buffer	–	–	–	–	–	10	10	–	–
fifo_samples_per_label	–	–	–	–	–	–	–	10	10
alpha	–	–	–	–	–	–	0.5	–	–
beta	–	–	–	–	–	–	0.5	–	–

Figure 16: Tiny ImageNet model-specific hyperparameters for all methods.

D.5 ImageNet - 1000 Model Specifications

Parameter	BP	CBP	EWC	Regen	Concat	ER	DarkExp	Full-Attn	QOA
step_size	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4
momentum	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
weight_decay	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
hidden_layer_size	100	100	–	–	100	100	100	100	50
optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
reset_head	–	false	–	–	–	–	–	–	–
early_stopping	–	true	–	–	–	–	–	–	–
replacement_rate	–	1e-5	–	–	–	–	–	–	–
utility_function	–	contribution	–	–	–	–	–	–	–
maturity_threshold	–	1000	–	–	–	–	–	–	–
noise_std	–	0	–	–	–	–	–	–	–
perturb_weights	–	false	–	–	–	–	–	–	–
ewc_lambda	–	–	1000	–	–	–	–	–	–
regenerative_lambda	–	–	–	1e-5	–	–	–	–	–
buffer_size	–	–	–	–	–	2000	2000	100	100
samples_from_buffer	–	–	–	–	–	10	10	–	–
fifo_samples_per_label	–	–	–	–	–	–	–	10	10
alpha	–	–	–	–	–	–	0.5	–	–
beta	–	–	–	–	–	–	0.5	–	–

Figure 17: ImageNet - 1000 model-specific hyperparameters for all methods.

D.5.1 Regression (SCR): Model Specifications

Parameter	BP	CBP	EWC	Regen	Concat	ER	DarkExp	Full-Attn	QOA
num_inputs	20	20	20	20	20	20	20	20	41
num_outputs	1	1	1	1	1	1	1	1	1
num_features	40	40	40	40	40	40	40	14	20
activation	ReLU	ReLU	ReLU	ReLU	ReLU	ReLU	ReLU	ReLU	ReLU
step_size	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4
weight_decay	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
β_1	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
β_2	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
ewc_lambda	–	–	100000	–	–	–	–	–	–
regenerative_lambda	–	–	–	0.01	–	–	–	–	–
buffer_size	–	–	–	–	–	2000	2000	–	–
fifo_buffer_size	–	–	–	–	–	–	–	100	100
samples_from_buffer	–	–	–	–	–	1	1	–	–
fifo_samples	–	–	–	–	–	–	–	100	100
alpha	–	–	–	–	–	–	0.5	–	–
beta	–	–	–	–	–	–	0.5	–	–

Figure 18: Model hyperparameters for regression experiments (SCR). All models use the Adam optimizer with shared base settings. Method-specific parameters are listed where applicable.

D.6 Permuted MNIST Setup

The MNIST dataset [24] consists of 60,000 training and 10,000 test images of hand-written digits (0–9), each represented as a 28×28 grayscale image. To adapt this dataset for continual learning, we make the following modifications:

- **Train/test split.** For each task, we use the entire 60,000 original training images. These are further divided into 58,000 images for training and 2,000 images for evaluation within the task. The global MNIST test set is not used directly; instead, we re-sample 2,000 held-out examples per task to serve as test data. This ensures consistency across tasks and keeps evaluation lightweight.

- **Downsampling.** To reduce computational cost, all images are downsampled from 28×28 to 7×7 , giving 49 input features per image.
- **Task generation.** Each task corresponds to a new random permutation of the 49 input pixels. The same permutation is applied consistently to all 60,000 images within a task. Labels remain unchanged.
- **Continual stream.** The learner observes tasks sequentially. After completing all 58,000 training pairs of a given permutation, the learner encounters remaining 2000 data-points for testing, following which it encounters the next task (with a new permutation). In total, 7,000 such tasks are generated in the continual stream.

This setup forces the continual learner to adapt to a new input representation (permutation) at the start of each task, while retaining performance on past permutations. It is a widely used benchmark for evaluating both *plasticity* (ability to adapt to new tasks) and *stability* (ability to avoid catastrophic forgetting).

Table 4: Permuted MNIST configuration: Query-Only Attention (V1).

Setting	Value
Support size ($ B $)	1000
Replay buffer size	1000
Distance metric	Learned Q_θ (query-only)
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0.0
Batching	(batch size = 400)
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	7500
Steps per task	150
(Input size, Hidden Size, Output Size, Hidden Layers)	(108, 100, 1, 9)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	6.5 hours/run

Table 5: Permuted MNIST configuration: Query-Only Attention (V2).

Setting	Value
Support size ($ B $)	200
Replay buffer size	200
Distance metric	Learned Q_θ (query-only)
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0.0
Batching	batch size = 400
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	7500
Steps per task	150
(Input size, Hidden Size, Output Size, Hidden Layers)	(108, 100, 1, 9)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	4 hours/run

Table 6: Permuted MNIST configuration: MAML.

Setting	Value
(Support size, query size, tasks per iteration) ($ B $)	(10, 10, 5)
Replay buffer size	50000
Optimizer	Adam
Outer Learning rate	$1e-4$
Inner Learning rate	$1e-2$
Weight decay	0.0
Batching	batch size = 400
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	75
Steps per task	150
(Input size, Hidden Size, Output Size, Hidden Layers) :	(49, 100, 10, 3)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	4.2 hours

The attention network for continual learning is directly adapted from [36], which contains the comprehensive details.

Table 7: Permuted MNIST configuration: Attention Network baseline.

Setting	Value
Support size ($ B $)	100
Attention	Full self-attention ($\mathcal{O}(n^2)$)
Replay buffer size	100
Optimizer	Adam
Learning rate	$5e-4$
Weight decay	0.0
Batching	batch size = 400
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	7500
Steps per task	150
Architecture	10 layers, 1 head, d_model=59
(Attention Layers, Attention Heads, Dimension) :	(10, 1, 59)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	10 hours/run

The vanilla backpropagation and continual backpropagation algorithm is from paper [12], which contains more details.

Table 8: Permuted MNIST configuration: Vanilla Backpropagation.

Setting	Value
Support size	N/A
Replay buffer size	0 (no replay)
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0.0
Batching	Online (batch size = 1)
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	7500
Steps per task	150
(Input size, Hidden Size, Output Size, Hidden Layers) :	(49, 200, 10, 3)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	5.0 hours/run

Table 9: Permuted MNIST configuration: Continuous Backpropagation.

Setting	Value
Support size	N/A
Replay buffer size	0 (no replay)
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0
Batching	Online (batch size = 1)
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	7500
Steps per task	150
(Input size, Hidden Size, Output Size, Hidden Layers) :	(49, 200, 10, 3)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	7.0 hours/run

D.7 Split Image Net

We adopt the Split Image Net benchmark introduced by Dohare et al. [12]. However, due to computational constraints we utilize tiny version of the image net instead of full image net. We further downsample the images to $32 * 32$ for faster computation. The remaining setup is same as in the original paper, ensuring comparability of results. For full details, we refer readers to the experimental protocol in [12]. Tiny image net consists of 200 labels and 500 images per label in training setup. The test setup consists of 50 labels per class. To incorporate tiny image net for continual learning, each task consists of randomly sampling images from 2 classes. Thus a task is a binary classification task with 1000 datapoints for training. At the end of training in that class we measure the accuracy on 100 datapoints corresponding to 2 labels used in training. Since the training is purely online fashion a task is trained for a single epoch and then validation is carried out, followed by images for next task. All the below architectures presented use the same CNN architecture as the starting point for input image transformation:

Table 10: Split Image Net: CNN architecture (same for all models)

Setting	Value
(Input channels, Output Channels, Kernel Size, Padding, MaxPool, Activation)	(3, 32, 5, 1, 2, Relu)
(Input channels, Output Channels, Kernel Size, Padding, MaxPool, Activation)	(32, 64, 3, 1, 2, Relu)
(Input channels, Output Channels, Kernel Size, Padding, MaxPool, Activation)	(64, 128, 3, 1, 2, Relu)
Init	Xavier uniform (weights), Zeros (biases)

Table 11: Split Image Net: Query-Only Attention.

Setting	Value
Support size ($ B $)	180
Replay buffer size	200
Distance metric	Learned Q_θ (query-only)
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0.0
Batching	(batch size = 10)
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	9500
Steps per task	150
(Input size, Hidden Size, Output Size, Hidden Layers)	(2304, 128, 1, 1)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	$1 \times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	3 hours/run

Table 12: Split Image Net: MAML.

Setting	Value
(Support size, query size, tasks per iteration) ($ B $)	(10, 10, 5)
Replay buffer size	20000
Optimizer	Adam
Outer Learning rate	$1e-4$
Inner Learning rate	$1e-2$
Weight decay	0.0
Batching	batch size = 10
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	500
Steps per task	150
(Input size, Hidden Size, Output Size, Hidden Layers) :	(1152, 128, 2, 1)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	$1 \times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	5 hours (ran only first 100 tasks)

Table 13: Split Image Net: Attention Network baseline.

Setting	Value
Support size ($ B $)	180
Attention	Full self-attention ($\mathcal{O}(n^2)$)
Replay buffer size	200
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0.0
Batching	batch size = 10
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	9500
Steps per task	150
Architecture	3 layers, 1 head, d_model=130
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	4.5 hours/run

The vanilla backpropagation and continual backpropagation algorithm is from paper [12], which contains more details.

Table 14: Split Image Net: Vanilla Backpropagation.

Setting	Value
Support size	N/A
Replay buffer size	0 (no replay)
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0.0
Batching	Online (batch size = 10)
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	9500
Steps per task	150
(Input size, Hidden Size, Output Size, Hidden Layers) :	(1152, 128, 2, 1)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	4.0 hours/run

Table 15: Split Image Net: Continuous Backpropagation.

Setting	Value
Support size	N/A
Replay buffer size	0 (no replay)
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0
Batching	Online (batch size = 1)
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	7000
Steps per task	150
(Input size, Hidden Size, Output Size, Hidden Layers) :	(1152, 128, 2, 1)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	5.5 hours/run

Table 16: EWC: Elastic Weight Consolidation.

Setting	Value
Lambda	100000
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0
Batching	Online (batch size = 1)
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	7000
Steps per task	150
(Input size, Hidden Size, Output Size, Hidden Layers) :	(1152, 128, 2, 1)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	5.5 hours/run

D.8 Slowly changing regression (SCR)

We adopt the Slowly changing regression (SCR) benchmark introduced by Dohare et al. [12], which was designed to study loss of plasticity in continual learning. In this task, regression targets evolve gradually over time according to smoothly drifting functions, creating a non-stationary data stream. We use the same setup and data-generation procedure as in the original paper, ensuring comparability of results. For full details, we refer readers to the experimental protocol in [12].

Table 17: Slowly changing regression task: Query-Only Attention.

Setting	Value
Support size ($ B $)	100
Replay buffer size	100
Distance metric	Learned Q_θ (query-only)
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0.0
Batching	batch size = 1
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	800
Steps per task	10000
(Input size, Hidden Size, Output Size, Hidden Layers) :	(40, 20, 1, 1)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	7 hours/run

Table 18: Slowly changing regression task: MAML.

Setting	Value
(Support size, query size, tasks per iteration) ($ B $)	(10, 10, 5)
Replay buffer size	20000
Optimizer	Adam
Inner Learning rate	$1e-2$
Outer Learning rate	$1e-4$
Weight decay	0.0
Batching	batch size = 1
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	800
Steps per task	10000
(Input size, Hidden Size, Output Size, Hidden Layers) :	(20, 40, 1, 1)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	8.5 hour to run 100 tasks

Table 19: Slowly changing regression task: Attention Network.

Setting	Value
Support size ($ B $)	100
Replay buffer size	100
Distance metric	Learned Q_θ (query-only)
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0.0
Batching	batch size = 1
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	800
Steps per task	10000
(Attention Layers, Attention Heads, Dimension) :	(1, 1, 21)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	14 hours/run

Table 20: Slowly changing regression task: Vanilla Backpropagation

Setting	Value
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0.0
Batching	batch size = 1
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	800
Steps per task	10000
(Input size, Hidden Size, Output Size, Hidden Layers) :	(20, 40, 1, 1)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	8 hours/run

Table 21: Slowly changing regression task: Continuous Backpropagation

Setting	Value
Optimizer	Adam
Learning rate	$1e-4$
Weight decay	0.0
Batching	batch size = 1
Seeds	20, 30, 40 (report mean $\pm$ std)
Tasks	800
Steps per task	10000
(Input size, Hidden Size, Output Size, Hidden Layers) :	(20, 40, 1, 1)
Init	Xavier uniform (weights), Zeros (biases)
Hardware	1 $\times$ RTX 3090 24GB, CUDA 11.8
Wall-clock	11 hours/run

NeurIPS Paper Checklist

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: The paper explicitly acknowledges limitations, including the reliance on a support set (which may be restrictive in some continual learning scenarios), limited evaluation on permuted MNIST , Image Net and regression tasks, and the need for more rigorous theoretical analysis. We also note that computational cost and scalability of support-based methods remain open challenges for future work. Please refer to section 6 and A.2.

3. Theory assumptions and proofs

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Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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Justification: We have provided link to GitHub URL for, ensuring reproducibility while preserving double-blind review. This will allow faithful reproduction of the main results. The GitHub is code currently being cleaned for easy reproducibility. Please refer to section D.6 , D.7 and D.8 for reproducibility.

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Possible negative impacts include misuse in surveillance or adaptive profiling systems, where continual learning might enable intrusive or biased tracking over time. However, since our work focuses on simplified models (query-only attention) and theoretical analysis, these risks are indirect.

We explicitly acknowledge these as broader considerations, and stress that the current work is a step toward understanding fundamental mechanisms (plasticity vs. forgetting), not a directly deployable system. Please refer to section [A.1](#)

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Justification: We use only publicly available benchmark datasets and assets, including MNIST [\[24\]](#) (permuted variant), tiny image net and synthetic regression tasks (SCR). MNIST is distributed under a permissive license for research use, and we cite the original source. We also cite all baseline algorithms (e.g. transformers [\[34\]](#), CBP [\[12\]](#)). No proprietary or restricted data, models, or code were used, and all assets are properly credited and used within their intended terms.

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Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

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